



Final Project — Build Your Own CRUD App

The goal: take everything from the 3 sessions and build *your own* mobile app, backed by a real **SQLite** database, with full **CRUD**.

You already have every skill you need. The final project is the **same patterns** you practiced — just on **data you care about**.

Suggested time: 1–2 weeks · **Deliverable:** a working app you demo on your phone + the code.

1. Pick an idea

Choose one (or pitch your own). For each, here's a starter **table design** — that's the first thing you should plan.



Task / To-Do Manager

Build on what you made. Add categories or due dates.

```
tasks( id, title, category TEXT, due TEXT, done INTEGER )
```



Expense Tracker

Log spending and show a running total.

```
expenses( id, title, amount REAL, category TEXT, date TEXT )
```



Notes / Journal

Write, edit, and search notes.

```
notes( id, title, body TEXT, created TEXT )
```

Contacts Book

Save people; tap to view/edit.

```
contacts( id, name, phone TEXT, email TEXT )
```

Inventory / Stock Tracker

Track items and quantities (+/-).

```
items( id, name, quantity INTEGER, price REAL )
```

Habit Tracker

Check off daily habits, count a streak.

```
habits( id, name, streak INTEGER, lastDone TEXT )
```

 **Make it personal.** Track *your* expenses, *your* study habits, *your* book list. You'll enjoy it more and build it better.

2. Requirements (the MVP — required for full marks)

Your app **must**:

1.  Use a **SQLite table** with **at least 3 columns**
2.  Do **all four CRUD** operations:
 - **Create** — add new records (**INSERT**)
 - **Read** — list/show records (**SELECT**)
 - **Update** — edit an existing record (**UPDATE**)
 - **Delete** — remove a record (**DELETE**)
3.  **Persist** — data survives a full app restart
4.  Have a **list screen** + an **add/edit form** (a **Modal** is fine)
5.  Show an **empty state** when there's no data
6.  **Confirm before deleting** (**Alert**)

7.  Be **styled** consistently (colors, spacing, rounded cards)
8.  Use **parameterized queries** () — never glue strings into SQL

3. Stretch goals (the extra 10%)

Pick a few to stand out:

-  **Search / filter** the list
-  **Sort** (by date, name, amount, done)
-  **Categories** or tags
-  A **summary** (total spent, tasks left, longest streak)
-  **Real navigation** with **Expo Router** (separate screens instead of a Modal)
-  **Dark mode** or a theme switcher
-  A custom **app icon** & splash screen
-  A **date picker** (`@react-native-community/datetimepicker`)

4. Milestones

Milestone	When	What's done
M1 — Plan	Day 1–2	Idea chosen · table schema designed · screens sketched on paper
M2 — Create + Read	Week 1	<code>db.js</code> + table · add records · list them · persists on restart
M3 — Update + Delete	Week 1–2	Edit form (Modal) · delete with confirm · all CRUD works
M4 — Polish	Final	Empty states · styling · 1–2 stretch goals · demo-ready

Build in this order. Get **Create + Read persisting first** — that's the hard part, and everything else is quick once it works.

5. How you'll be graded (100 pts)

Area	Pts	What we look for
Functionality & CRUD	40	All four operations work; data persists across a restart
Data model & persistence	20	Sensible table/columns; SQLite used correctly; <code>?</code> placeholders
UI / UX & styling	20	Clean, consistent design; empty state; confirm-delete; readable
Code quality & structure	10	DB code in <code>db.js</code> ; reusable components; tidy, readable code
Creativity & stretch	10	Useful idea; at least one stretch feature; polish
Live demo	—	You run it on a real phone and walk us through CRUD

A passing project: adds, lists, edits, and deletes records that survive a restart, looks tidy, and runs on a phone.

6. Submitting

1. **Code:** push to a GitHub repo (public), **or** share an [Expo Snack](#) link, **or** a zip (without `node_modules`).
2. **README:** 4–5 lines — what the app does, your table schema, and how to run it (`npm install` → `npx expo start`).
3. **Demo:** a ~2-minute walkthrough on your phone (live in class, or a screen recording).

7. Tips for success 🌟

- **Start from the `session-3` branch** — it already has working CRUD + a Modal. Rename `tasks` to your thing and change the columns. *Don't start from a blank file.*
- **Change** → `refresh()` after every database write. This one habit prevents most bugs.
- **One feature at a time.** Get adding working before editing; get the data right before the styling.
- **Test persistence early and often** — force-close and reopen.
- **Stuck?** Re-read the Session 3 slides and presenter code. The answer is almost always there.

You went from *never coding* to *shipping a database-backed mobile app* in three weeks. The final project is your victory lap — make something **you'd actually use**. 🚀